Machine Learning

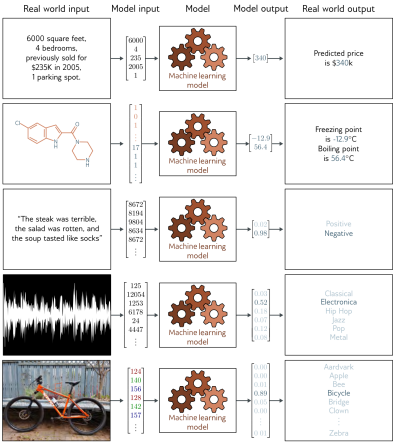
3. Ensemble Learning, Regularization; ML challenges

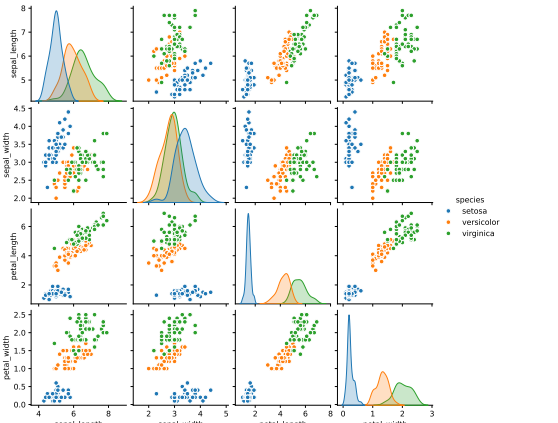
Louis Bagot
Université Claude Bernard Lyon 1

October 19, 2026

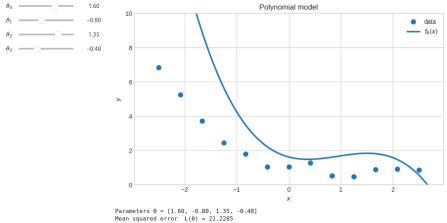
Where were we?

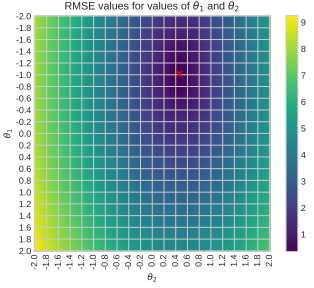
ML: Approximate a function from data examples $\{x_i, y_i\}$





Parametrized models $f_\theta(x_i) \approx y_i$ learn parameters θ to minimize loss: $\arg\min_\theta \mathcal{L}(\mathcal{D}, f_\theta)$





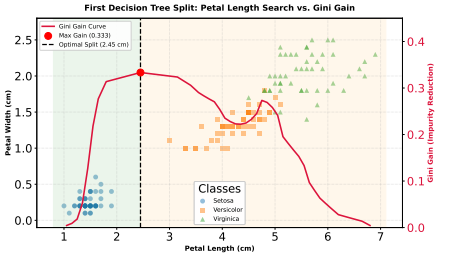
Where were we?
Adapting models & metrics for Classification

- Use a **softmax** to convert generic logits to probabilities
- Use the **CrossEntropy** loss with softmax to strongly punish errors
- Use **accuracy**, **recall**, **precision** to evaluate classif

Where were we?
Decision Trees

Recursively **split** the space along features:

- list all possible splits
  (interstices between datapoints)
- evaluate split quality
  (improvement in metric: Gini, MSE)
- take the best split; repeat



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Imagine that we can use a bunch of those...

Can we combine their strengths?

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Most competition-winning methods have been based on Ensembles.

- critical question to keep in mind:
  why use M models instead of a single, M-times bigger model?

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How do we train and use those?

- $\rightarrow$ what input, output, model, loss function?
- $\rightrightarrows$ we will see *stacking*, *bagging* and *boosting*!

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Simplest solution!

$$f(\boldsymbol{x}) = \frac{1}{M} \sum_{m=1}^{M} f_m(\boldsymbol{x})$$

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Potential issues:

- not all the same quality
  - $\rightarrow$ *weigh* them? $w_m = 0$ removes bad models:

$$f(\boldsymbol{x}) = \sum_{m=1}^{M} w_m f_m(\boldsymbol{x})$$

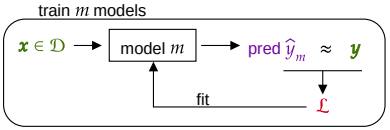
this is just a linear regression!

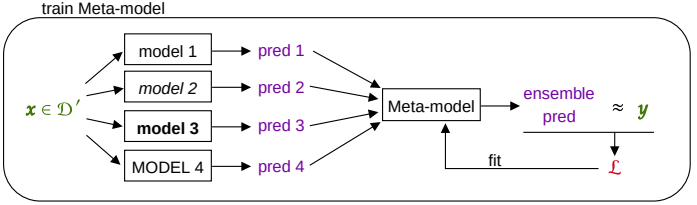
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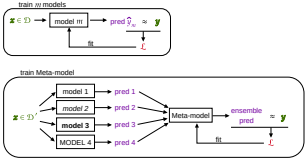
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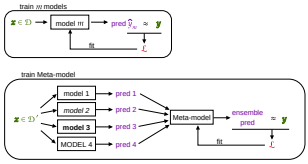


When does Stacking work?



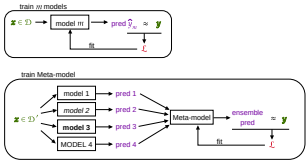
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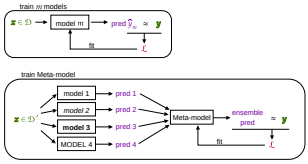
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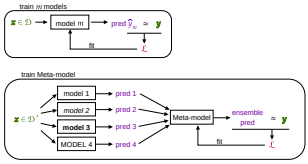
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- Note, in the graphic, $\mathcal{D}$ and $\mathcal{D}'$:
  - **do not** train models & Meta on the same data!

  $\rightarrow$ otherwise it will $\arg\max$!



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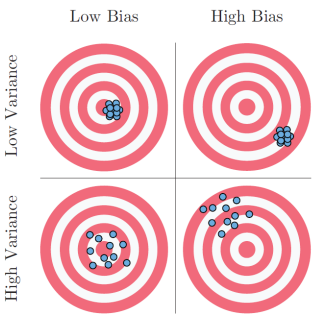
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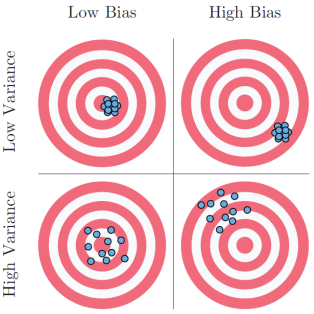
- In average, does my model find the solution?
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- How different are solutions found?
  - → Notion of **variance**
    - if my model overfits, lots of variance!
    - also if sensitive to noise, outliers, unstable



Why not use a single, M-times bigger model?

Notion of **bias-variance trade-off**:

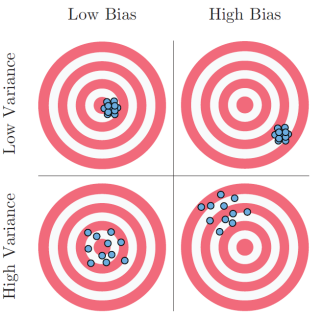
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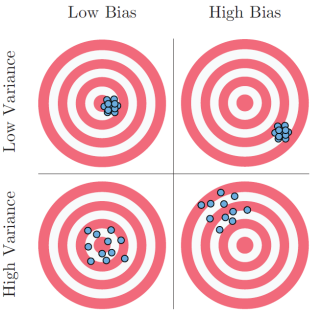


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→ **Stacking reduces variance**

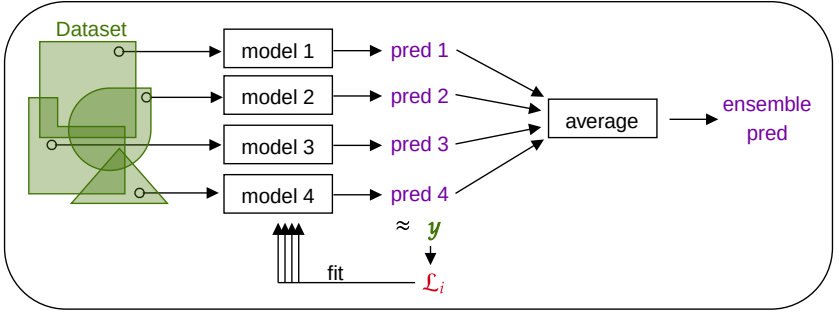


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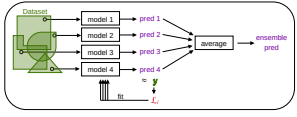
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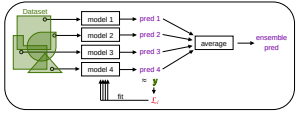


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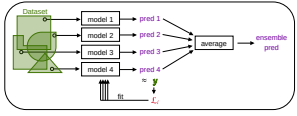


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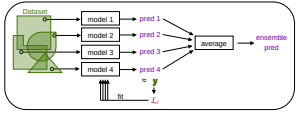


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- Sub-datasets sampled through *bootstrapping*: sample N times with replacement
  - $\rightarrow$ simulating different datasets!



What model family to use for bagging?

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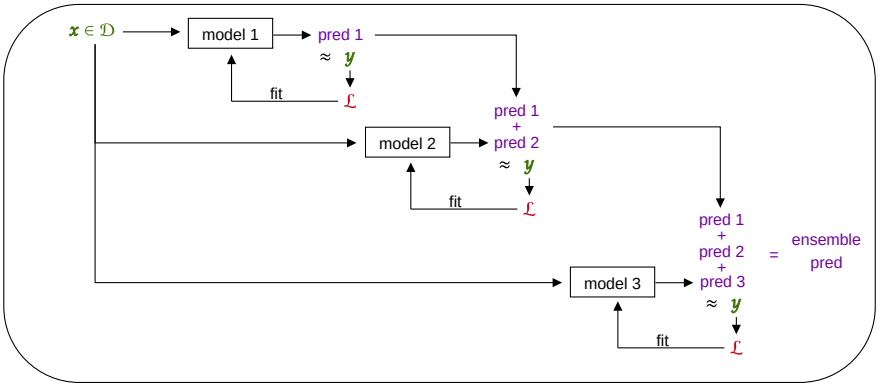
→ enhanced Bagging idea by also taking random features to split over

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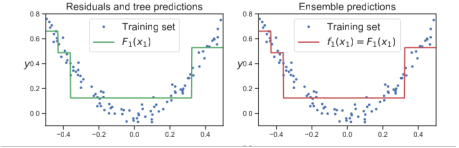


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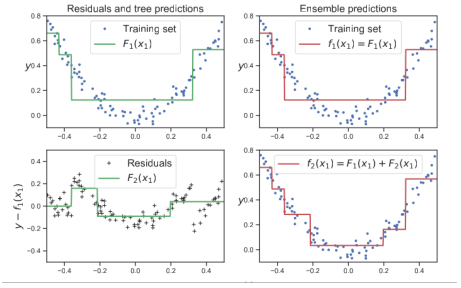


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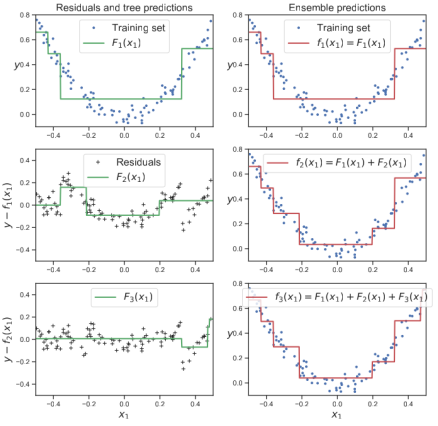


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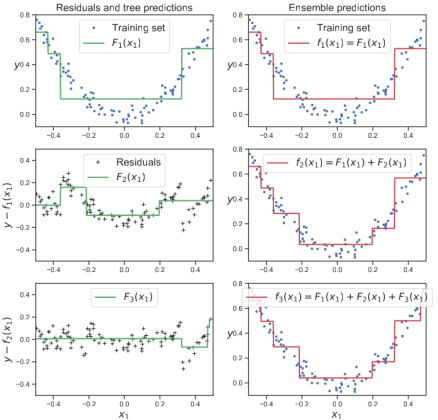


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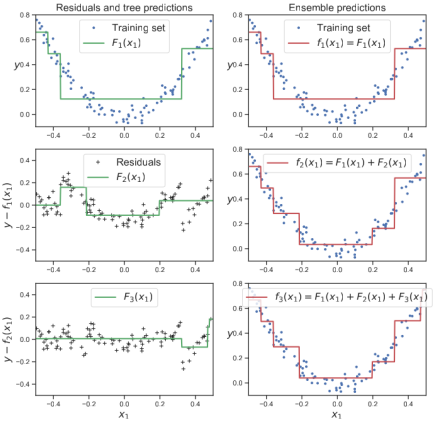
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$$f(\boldsymbol{x}) = \sum_{m=1}^{M} f_m(\boldsymbol{x})$$



Boosting: how does it behave?

Stacking and Bagging train independent models in parallel.

Boosting trains models **sequentially**!

- Not as computationally efficient
- Boosting **reduces bias** more than variance
  - often stronger than random forests for trees
  - more susceptible to overfit!
- see the notebook "`tree_vs_ensemble`" to play with it

Adding predictions for regression: OK. Classif though?

- If I use sigmoid$(f_1(\boldsymbol{x})) \approx \boldsymbol{y}$, how do I train f_2?
  - $\rightarrow$ I'd like my final prediction to be sigmoid$(f_1(\boldsymbol{x}) + f_2(\boldsymbol{x})) \approx \boldsymbol{y}$...
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- Observation: $f_2(\boldsymbol{x}) \approx \boldsymbol{y} - f_1(\boldsymbol{x})$ is the **negative gradient** of MSE!

$$-\frac{\partial(\boldsymbol{y} - f_1(\boldsymbol{x}))^2}{\partial f_1(\boldsymbol{x})} = \boldsymbol{y} - f_1(\boldsymbol{x}) \approx f_2(\boldsymbol{x})$$

  - $\rightarrow$ answers "how should I change $f_1(\boldsymbol{x})$ to be MSE-closer to $\boldsymbol{y}$? "
  - $\rightrightarrows$ great for regression with MSE! ... not others!

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Through their gradients,
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Name	Loss	$-\partial\ell(y_i, f(\boldsymbol{x}_i))/\partial f(\boldsymbol{x}_i)$
Squared error	$\frac{1}{2}(y_i - f(\boldsymbol{x}_i))^2$	$y_i - f(\boldsymbol{x}_i)$
Absolute error	$\lvert y_i - f(\boldsymbol{x}_i)\rvert$	$\operatorname{sgn}(y_i - f(\boldsymbol{x}_i))$
Exponential loss	$\exp(-\tilde{y}_i f(\boldsymbol{x}_i))$	$-\tilde{y}_i \exp(-\tilde{y}_i f(\boldsymbol{x}_i))$
Binary Logloss	$\log(1 + e^{-\tilde{y}_i f_i})$	$y_i - \pi_i$
Multiclass logloss	$-\sum_c y_{ic} \log \pi_{ic}$	$y_{ic} - \pi_{ic}$

Table 18.1: Some commonly used loss functions and their gradients. For binary classification problems, we assume $\tilde{y}_i \in \{-1, +1\}$, and $\pi_i = \sigma(2f(\boldsymbol{x}_i))$. For regression problems, we assume $y_i \in \mathbb{R}$. Adapted from [HTF09, p360] and [BH07, p483].

(you don't need to know the losses and grads, just the idea)

(libraries implement this for you!)

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Regularization: any process that forces the model to find simpler solutions.

- **The** main ML approach to fighting overfit
- Can be seen as trading some bias for variance
- Main method: **add a penalty term to the loss/objective** for complexity

Regularization: what penalty term to use?

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General observation: fitting specific datapoints requires **high weights/values**

$\rightarrow$ penalize high weights by adding their magnitude to the loss!

$$\arg\min_{\boldsymbol{\theta}} \mathcal{L}(\mathcal{D}, f_{\boldsymbol{\theta}}) + \lambda \sum_{i=1}^{\text{n_param}} {\theta_i}^2$$

$\rightrightarrows$ each weight θ_i is encouraged to be small

- λ controls the balance "solving the objective $\leftrightarrow$ model complexity"

Regularization: what penalty term to use?

- $\sum_{i=1}^{\text{n_param}} {\theta_i}^2 \doteq \|\boldsymbol{\theta}\|_2^2$ – This is the L2 norm of vector $\boldsymbol{\theta}$:
  - this is **L2 regularization** or **Ridge regression** (for linear models)
  - more strongly penalizes higher parameters; difficult to reach 0 exactly

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- "elastic net" regularization does L1+L2
- **!! magnitude of linear weights depends on the magnitude of variables !!**
  - → **normalize**!!

Ridge/Lasso: why use Regularization for linear models?

Since they don’t really overfit..?

Ridge/Lasso: why use Regularization for linear models?

Since they don't really overfit..?

- They **can** overfit! Depends on the features (see next slide!)
- Regularization can help **remove useless features**:
  - if two features are strongly correlated (e.g., weight and #items or cm and inch): force one of them down
  - if a feature is useless: lasso can make its weight exactly 0
    - $\rightarrow$ form of **feature selection**!

Regularization: what does it look like?

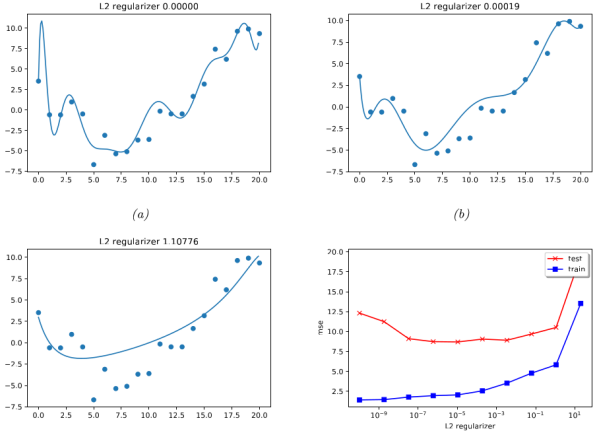


Figure 4.5: (a-c) Ridge regression applied to a degree 14 polynomial fit to 21 datapoints. (d) MSE vs strength of regularizer. The degree of regularization increases from left to right, so model complexity decreases from left to right. Generated by linreg_poly_ridge.ipynb.

Regularization: what about Decision Trees?

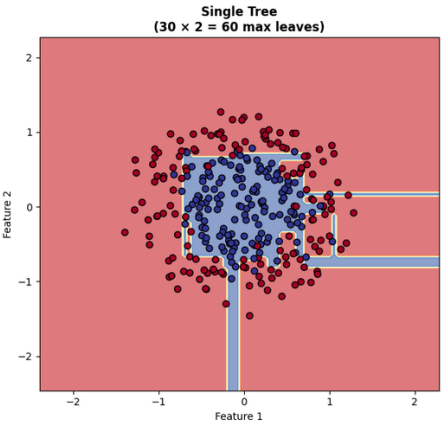
Not much sense to punish the split's feature value or the leaf value...

Regularization: what about Decision Trees?

Not much sense to punish the split's feature value or the leaf value...

Tree complexity controlled by
number of leaves and tree depth

- → penalize those!
- ⇉ prevent splits with too little perf increase



Regularization: what about Decision Trees?

In the context of Tree Gradient Boosting specifically

- consider each added tree's node as a "small correction"
  - $\rightarrow$ we want to prevent huge, unjustified corrections
  - $\rightrightarrows$ leaf values can be dampened by a factor!

Regularization: what about Decision Trees?

In the context of Tree Gradient Boosting specifically

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Decision Trees + Gradient Boosting + Regularizations + learning rate + more
= XGBoost, clear contender for "the best"

- Won too many competitions!
- Method described in a paper, XGBoost: A Scalable Tree Boosting System
- Library maintained by the community: xgboost
- start from the mean, add a few (3-6) shallow trees

Summary, please?

- **Ensemble Learning** combines models.
  - *Stacking*: learn to combine predictions
    - ▶ independent, diverse models from different families
    - ● more problem-specific than the others, no general algo
  - *Bagging*: generate diverse datasets
    - ▶ independent models from the same family, diverse data
    - ▶ Random Forest as the main interesting version
    - → balances out the tree's instability: variance reduction
    - ● perf is OKay-ish
  - *Boosting*: keep adjusting model predictions
    - ● reduces bias; great perf
    - ● more prone to overfitting
    - ▶ XGBoost MVP

Summary, please?

- **Ensemble Learning** combines models.
  - *Stacking*: learn to combine predictions
  - *Bagging*: generate diverse datasets
  - *Boosting*: keep adjusting model predictions

- **Regularization** addresses overfit
  - Prevents models from finding complicated solutions
  - Generally a penalty term on the loss (weight size, number of leaves)
  - Critical from linear models to neural nets, through trees!

Regularization: how do I pick my loss weight λ?

We have to tune it...

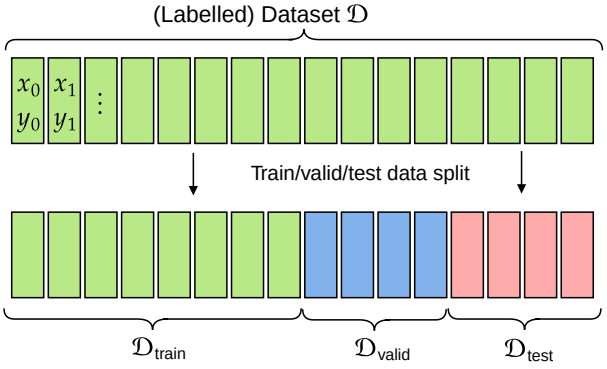
Hyper-parameter: any value that we pick and is not optimized by the model's algo

- KNN's k, number of neighbors
- Tree depth/max number of leaves
- Temperature for softmax (controls how "flat" it is)
- Number of models in Ensemble
- Regularization factors...
- ... and a lot more to come!
  - complicated methods tend to have more hyper-parameters...

How do I pick my hyper-parameters?

DO NOT optimize them against your test set!!

→ you're just going to find the best value *for your test set*.
instead, further break down your dataset:



How do I pick my hyper-parameters?

- For each hyper-parameter choice, **train** on the train set
- **Optimize** your hyper-parameters on the validation set
- **Evaluate** your model on the test set

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$\rightarrow$ OK, but how do I "optimize"?

- Good old: try a bunch of intuitive values by hand
- Grid search: try all of them within a window, e.g., $\lambda \in [0, 0.5, 1, 2]$
  - (values to try) (number of hyper-parameters)
- Smarter optimizers: see the course on MLOps!

$\rightarrow$ no matter how you do it, it's a pain!

Anything more I need to know?

Machine Learning is huge; lots more

- models (genetic algos, SVM, KDE...)
- problems (imbalance, anomaly detection, feature extraction, explainability...)
- solutions!

$\rightarrow$ impossible to cover in 10 or 30 hours!

Anything more I need to know?

Machine Learning is huge; lots more

- models (genetic algos, SVM, KDE...)
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- solutions!

$\rightarrow$ impossible to cover in 10 or 30 hours!

Point of this course:

Give you the tools to understand any problem, any method.

- All methods have advantages/disadvantages
- Always ask "why not do this?" "how would I do it?"
  - $\rightarrow$ (that's why all titles have been questions :D)
  - — **trust** in your ability to dive in, read docs, understand

Where to next?

Deep Learning course!

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- *Neural Networks* are one specific parametrized model
- Works really well on huge amounts of hugely complicated data
  - image
  - text
  - sound...

Where to next?

Deep Learning course!

- *Neural Networks* are one specific parametrized model
- Works really well on huge amounts of hugely complicated data
  - image
  - text
  - sound...
- "Traditional" ML methods are still relevant.
  - Linear relationship: nothing beats a linear regression.
  - Decision Trees can be read & visualized
  - For data of smaller dimension, Random Forests / XGBoost still win vs NNs.